

Topic 2.4 - Exponential Function Manipulation

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

Manipulating exponential functions is not cosmetic: a useful form can reveal an initial value, a factor for a new time unit, or a graph transformation. Every rewrite should preserve the function and make one feature easier to interpret.

Learning targets

- Rewrite exponential expressions using exponent properties and common bases.
- Translate growth factors across equivalent input intervals using rational exponents.
- Explain how algebraic equivalence corresponds to shifts, stretches, and unchanged graphs.

Language and structure

Term	Meaning in this topic
equivalent form	A different expression that gives the same output for every input in the domain.
rational exponent	An exponent such as $1/12$ that represents a root and supports time-scale conversion.
common base	A shared base used to compare or combine exponential expressions.
input scaling	Replacing x by cx , which changes the factor associated with one unit of input.
horizontal shift	Replacing x by $x - h$; exponent laws can convert the shift into a vertical scale factor.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Without using a calculator, decide which of the following are the same function: $f(x) = 4^{x+1}$, $g(x) = 16 \cdot 4^{x-1}$, and $h(x) = 2^{2x+2}$. Justify by rewriting rather than checking a few values.

Concept 1: Exponent laws and common bases

Core idea

Use $b^{u+v} = b^u b^v$, $b^{u-v} = b^u / b^v$, and $(b^u)^v = b^{uv}$.

- Rewrite numbers such as 8, 16, and 32 as powers of 2 when a common base is helpful.
- Coefficients outside an exponential cannot generally be moved into the exponent unless they are themselves powers of the base.
- Check equivalence by simplifying both expressions to the same coefficient, base, and exponent.

Worked example - annotate the reasoning

Problem. Show that $12(4)^{x-2}$, $\frac{3}{4}(4)^x$, and $3(2)^{2x-2}$ are equivalent.

Model reasoning. $12(4)^{x-2} = 12(4^x/16) = \frac{3}{4}4^x$. Also $3(2)^{2x-2} = 3(2^{2x}/4) = \frac{3}{4}(4^x)$.

Check your understanding

Independent

Rewrite $5(8)^{x+1}$ in the form $A(2)^{3x}$.

Concept 2: Changing the time unit

Core idea

If a factor R applies over k input units, then the one-unit factor is $R^{1/k}$.

- Annual factor from monthly factor m : m^{12} .
- Monthly factor from annual factor A : $A^{1/12}$.
- Always attach the factor to its interval; 1.08 per year and 1.08 per month describe very different change.

Worked example - annotate the reasoning

Problem. A population grows by a factor of 1.60 every 5 years. Write a model using a one-year growth factor.

Model reasoning. If $P(t) = P_0 b^t$ with t in years, then $b^5 = 1.60$, so $b = 1.60^{1/5}$. Thus $P(t) = P_0(1.60^{1/5})^t = P_0(1.60)^{t/5}$.

Check your understanding

Independent

An account grows by 2% per month. Write the equivalent annual growth factor and annual percent increase.

Concept 3: Algebraic form and graph transformations

Core idea

A horizontal shift in an exponential can be rewritten as a vertical scaling because $b^{x-h} = b^{-h}b^x$.

- $ab^{x-h} = ab^{-h}b^x$; the graph is unchanged.
- Replacing x by cx changes the effective one-unit factor from b to b^c .
- Two expressions can describe the same graph while highlighting different parameters.

Worked example - annotate the reasoning

Problem. Rewrite $f(x) = 7(3)^{x-2} + 4$ in the form $A(3)^x + 4$ and explain the graph equivalence.

Model reasoning. $7(3)^{x-2} + 4 = \frac{7}{9}(3)^x + 4$. Both expressions produce the same graph; the first highlights a right shift of 2, while the second highlights a vertical scale coefficient $7/9$.

Check your understanding

Independent

Find c so that $2(5)^{3x} = 2(c)^x$ for all real x .

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Rewrite $9(27)^{x-1}$ using base 3 and simplify the coefficient.

Exit ticket 2

No calculator

A factor of 1.44 applies every 2 years. Find the annual factor.

Exit ticket 3

No calculator

Explain why $2^{x+3} \neq 2^x + 2^3$.
